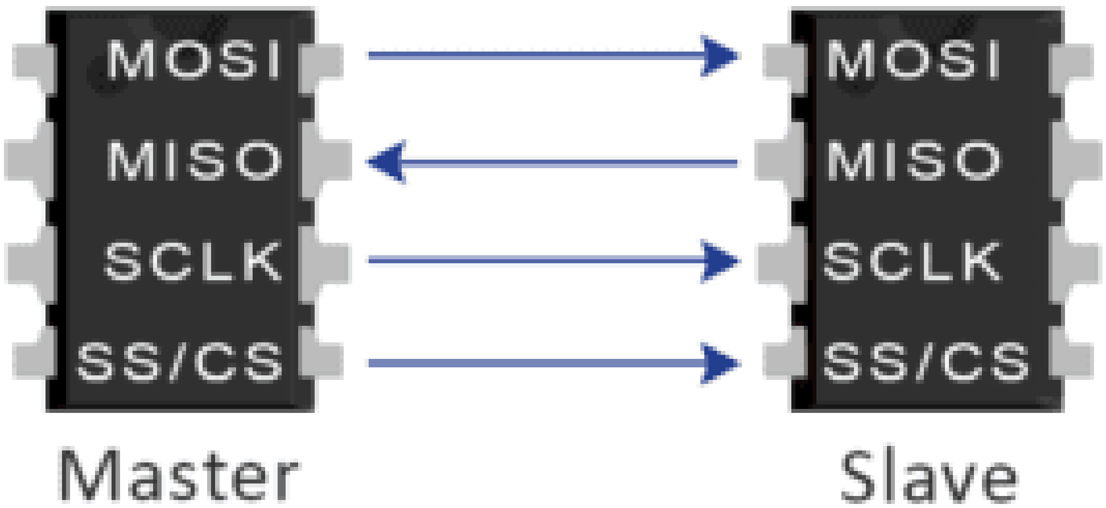
SPI Project



SPI Code:

```
module SPI (
    MOSI ,  SS_n , clk ,rst_n , tx_data , tx_valid , MISO , rx_valid , rx_data
);
input MOSI , SS_n , clk , rst_n , tx_valid ;
input [7:0] tx_data ;
output  [9:0] rx_data;
output reg MISO , rx_valid ;

// parameters
parameter IDLE      = 3'b000 ;
parameter CHK_CMD   = 3'b001 ;
parameter WRITE     = 3'b010 ;
parameter READ_ADD  = 3'b011 ;
parameter READ_DATA = 3'b100 ;

reg address_entered ; // make choice between read_data , read_add
reg [3:0] counter ;
reg [9:0] data_reg ;
```

```verilog
(* fsm_encoding = "one_hot" *)
reg [2:0] cs, ns;

//next state
always @(*) begin
    case (cs)

        IDLE: ns = (SS_n == 0) ? CHK_CMD : IDLE;

        CHK_CMD: begin
            if (SS_n == 1)
                ns = IDLE;
            else begin
                if (MOSI == 1) begin
                    if (address_entered == 1)
                        ns = READ_DATA;
                    else
                        ns = READ_ADD;
                end
                else
                    ns = WRITE;
            end
        end

        WRITE:
            ns = (SS_n == 1) ? IDLE : WRITE;

        READ_ADD:
            ns = (SS_n == 1) ? IDLE : READ_ADD;

        READ_DATA:
            ns = (SS_n == 1) ? IDLE : READ_DATA;

        default: ns = IDLE;

    endcase
end

// state memory
always @(posedge clk) begin
    if(~rst_n)
    cs <= IDLE ;
    else
    cs <= ns   ;
```

```
end

// Output
always @(posedge clk) begin


    if (~rst_n) begin
        MISO <= 0;

        rx_valid <= 0;
        counter <= 0;
        data_reg <= 0;
        address_entered <= 0;
    end

    else begin
        case(cs)


        IDLE: begin
            rx_valid <= 0;
        end


        CHK_CMD: begin
            counter <= 10;
        end


        WRITE: begin

            if(counter > 0) begin
                data_reg[counter-1] <= MOSI;
                counter <= counter - 1;
            end

            else begin
                rx_valid <= 1;

            end

        end


        READ_ADD: begin
```

```
        if(counter > 0) begin
            data_reg[counter-1] <= MOSI;
            counter <= counter - 1;
        end

        else begin
            rx_valid <= 1;
            address_entered <= 1;

        end

    end


    READ_DATA: begin

        if(tx_valid) begin

            rx_valid <= 0;

            if(counter > 0) begin
                  MISO <= tx_data[counter-1];
                counter <= counter - 1;
            end

            else begin

                 address_entered <= 0;
            end

        end

        else begin

            if(counter > 0) begin
                data_reg[counter-1] <= MOSI;
                counter <= counter - 1;
            end

            else begin
                rx_valid <= 1;

                counter <= 9;
            end
```

```
            end

        end


        default: begin
            rx_valid <= 0;
        end

        endcase
    end

end
assign rx_data = data_reg;

endmodule
```

RAM Code:

```
module RAM (
    din ,rst_n , clk , rx_valid , dout , tx_valid
);
parameter MEM_DEPTH = 256 ;
parameter ADDR_SIZE = 8   ;

input [9:0] din ;
input clk , rst_n , rx_valid ;

output reg tx_valid ;
output reg [ADDR_SIZE-1 : 0 ] dout ;



reg [ADDR_SIZE-1 : 0] wr_addr , rd_addr ;
reg [ADDR_SIZE-1 : 0] mem [MEM_DEPTH - 1 : 0] ;

always @(posedge clk) begin
    if (~rst_n)begin
        dout      <= 0 ;
```

```
        tx_valid  <= 0 ;
        wr_addr   <= 0 ;
        rd_addr   <= 0 ;
    end
    else begin
        if (rx_valid) begin
        case (din[9:8])

             2'b00: begin
               wr_addr <= din [ADDR_SIZE -1 : 0] ;
               tx_valid<= 0 ;
            end
             2'b01: begin
               mem [wr_addr] <= din [ADDR_SIZE -1 : 0] ;
               tx_valid<= 0 ;
            end
             2'b10: begin
               rd_addr <= din [ADDR_SIZE -1 : 0] ;
               tx_valid<= 0 ;
            end
             2'b11: begin
               dout    <= mem [rd_addr] ;
               tx_valid<= 1 ;
            end

        endcase
       end
    end
end
endmodule
```

Top module Code:

```
module SPI_RAM_TOP (
    input MOSI,
    input SS_n,
    input clk,
    input rst_n,
    output MISO
);

// internal wires
```

```verilog
wire [9:0] rx_data;
wire rx_valid;

wire [7:0] tx_data;
wire tx_valid;


// SPI instantiation
SPI SPI_inst (

    .MOSI(MOSI),
    .SS_n(SS_n),
    .clk(clk),
    .rst_n(rst_n),

    .tx_data(tx_data),
    .tx_valid(tx_valid),

    .MISO(MISO),

    .rx_valid(rx_valid),
    .rx_data(rx_data)

);


// RAM instantiation
RAM RAM_inst (

    .din(rx_data),
    .clk(clk),
    .rst_n(rst_n),

    .rx_valid(rx_valid),

    .dout(tx_data),
    .tx_valid(tx_valid)

);

endmodule
```

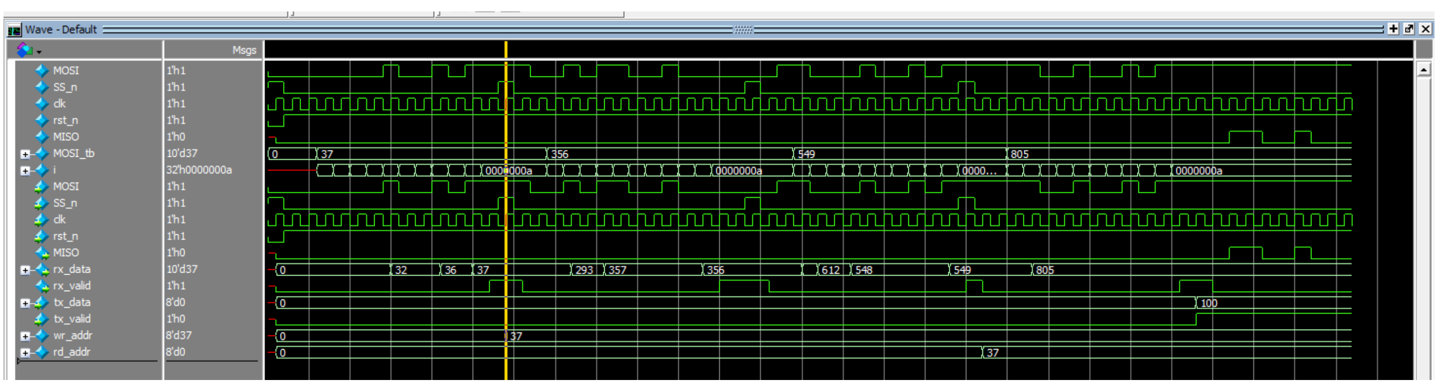
TestBench Code:

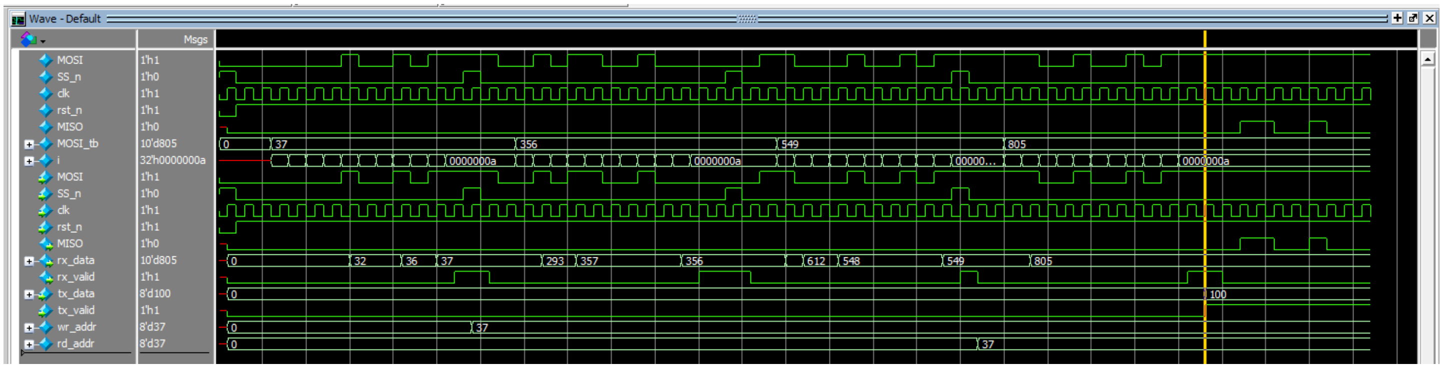
```
module SPI_tb ();
reg MOSI, SS_n, clk, rst_n;
wire MISO;

reg [9:0] MOSI_tb;

// DUT
SPI_RAM_TOP DUT (
    .MOSI(MOSI),
    .SS_n(SS_n),
    .clk(clk),
    .rst_n(rst_n),
    .MISO(MISO)
);

initial begin
    clk = 0;
    forever #1 clk = ~clk;
end

integer i;

initial begin
   $readmemh("mem.dat", DUT.RAM_inst.mem);

    rst_n = 0;
    MOSI = 0;
    SS_n = 1;
    MOSI_tb = 0;

    //Check Reset function
    @(negedge clk);
    if(MISO != 0) begin
        $display("Error");
        $stop;
    end

    //Check Write address function
    rst_n = 1;
    SS_n = 0;
    @(negedge clk);
    MOSI = 0;

    @(negedge clk);
    MOSI_tb = 10'b00_00100101 ; // 00 8'd37
```

```
for (i = 0 ; i < 10 ; i = i + 1 ) begin
    MOSI = MOSI_tb[9-i];
    @(negedge clk);
end

@(negedge clk);

//Check Write Data function
SS_n = 1;
@(negedge clk);
SS_n = 0;
@(negedge clk);
MOSI = 0;
@(negedge clk);

MOSI_tb = 10'b01_01100100 ; // mem[37] = 100

for ( i = 0 ; i < 10 ; i = i + 1 ) begin
    MOSI = MOSI_tb[9-i];
    @(negedge clk);
end

@(negedge clk);
@(negedge clk);

//Check read address in memory
SS_n = 1;
@(negedge clk);
SS_n = 0;
@(negedge clk);
MOSI = 1;
@(negedge clk);

MOSI_tb = 10'b10_00100101 ; //  addr_rd = 37

for ( i = 0 ; i < 10 ; i = i + 1 ) begin
    MOSI = MOSI_tb[9-i];
    @(negedge clk);
end

//Check read data from memory
SS_n = 1;
@(negedge clk);
SS_n = 0;
```

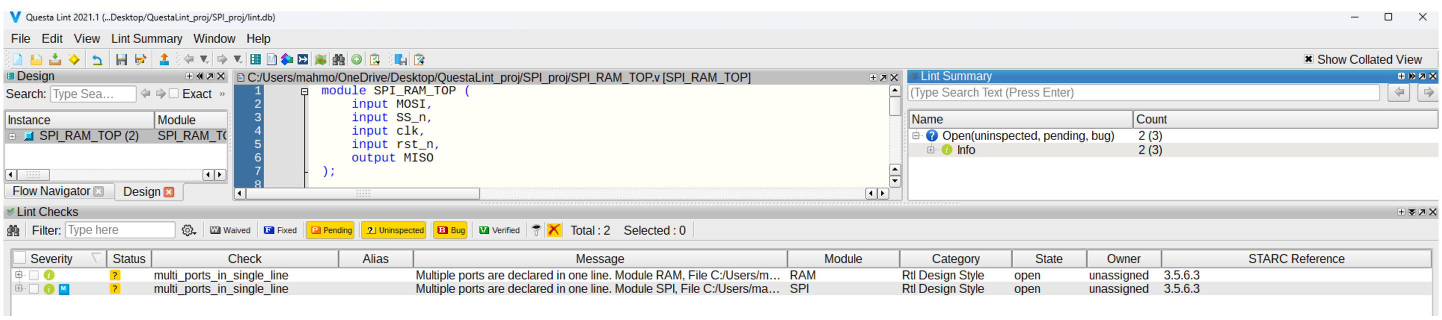
```
    @(negedge clk);
    MOSI = 1;
    @(negedge clk);

    MOSI_tb = 10'b11_00100101 ;

    for ( i = 0 ; i < 10 ; i = i + 1 ) begin
        MOSI = MOSI_tb[9-i];
        @(negedge clk);
    end

    repeat(11) @(negedge clk);

    $stop;
end

endmodule
```

QuestaSim:





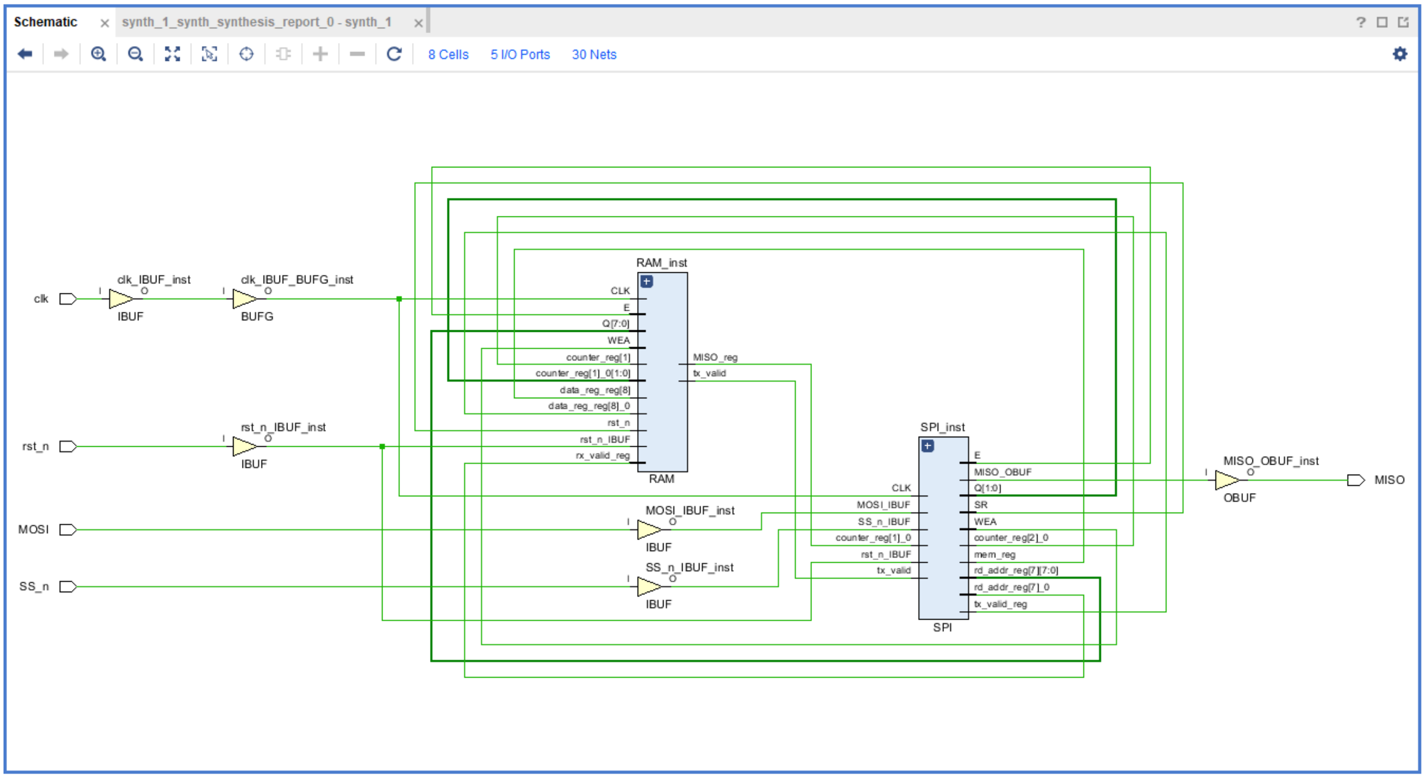
QuestaLint:



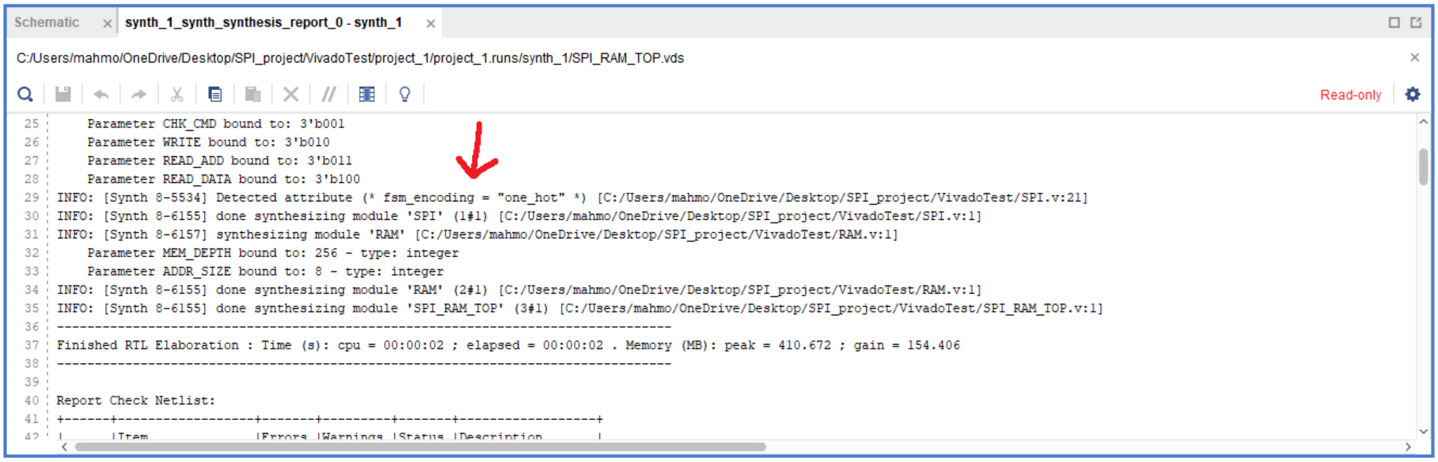
Vivado:-

One_hot:

Schematic:



Synthesis Report:



```
25      Parameter CHK_CMD bound to: 3'b001
26      Parameter WRITE bound to: 3'b010
27      Parameter READ_ADD bound to: 3'b011
28      Parameter READ_DATA bound to: 3'b100
29  INFO: [Synth 8-5534] Detected attribute (* fsm_encoding = "one_hot" *) [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/SPI.v:21]
30  INFO: [Synth 8-6155] done synthesizing module 'SPI' (1#1) [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/SPI.v:1]
31  INFO: [Synth 8-6157] synthesizing module 'RAM' [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/RAM.v:1]
32      Parameter MEM_DEPTH bound to: 256 - type: integer
33      Parameter ADDR_SIZE bound to: 8 - type: integer
34  INFO: [Synth 8-6155] done synthesizing module 'RAM' (2#1) [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/RAM.v:1]
35  INFO: [Synth 8-6155] done synthesizing module 'SPI_RAM_TOP' (3#1) [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/SPI_RAM_TOP.v:1]
36  ---------------------------------------------------------------------------------
37  Finished RTL Elaboration : Time (s): cpu = 00:00:02 ; elapsed = 00:00:02 . Memory (MB): peak = 410.672 ; gain = 154.406
38  ---------------------------------------------------------------------------------
39
40  Report Check Netlist:
41  +------+------------------+-------+---------+-------+------------------+
```

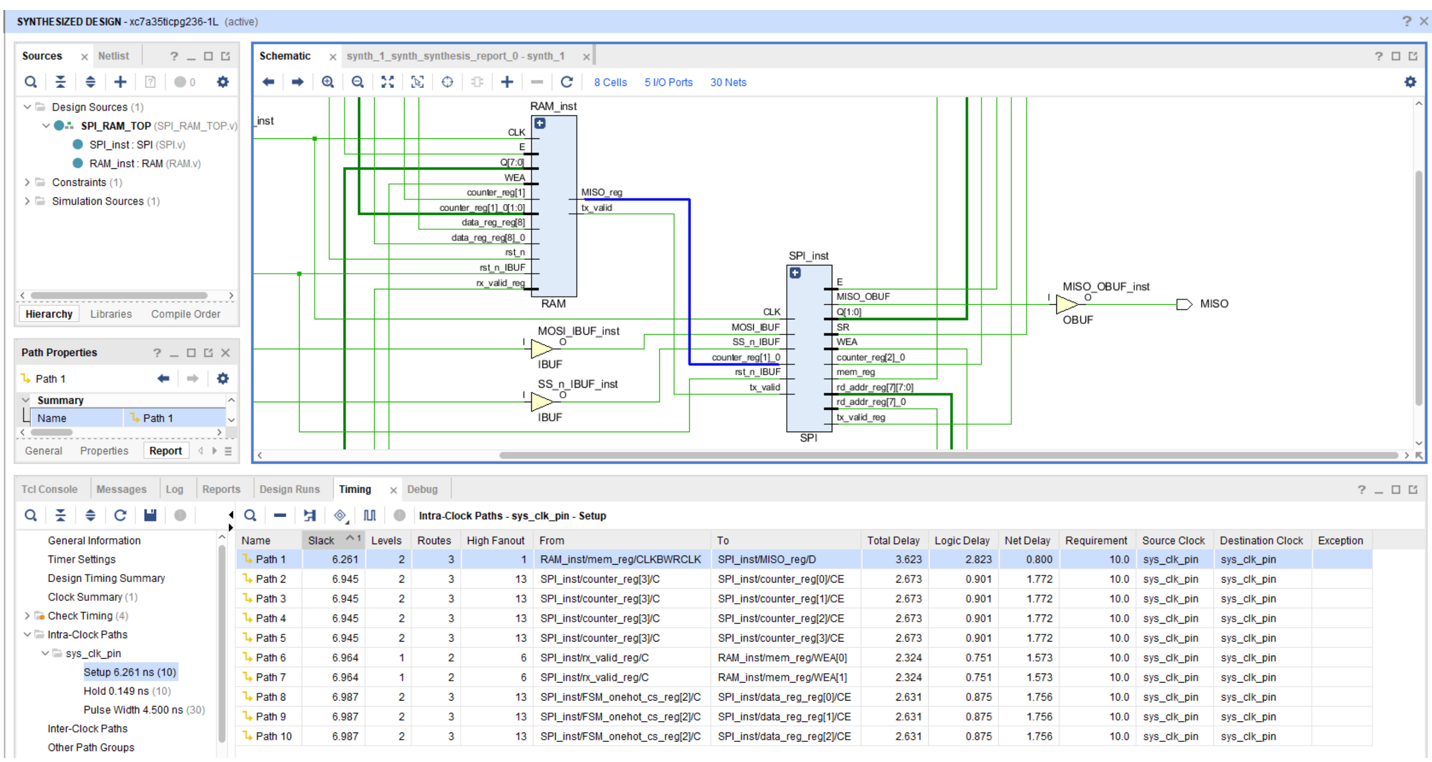

Timing Report:

Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	6.261 ns	Worst Hold Slack (WHS):	0.149 ns	Worst Pulse Width Slack (WPWS):	4.500 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	87	Total Number of Endpoints:	87	Total Number of Endpoints:	42

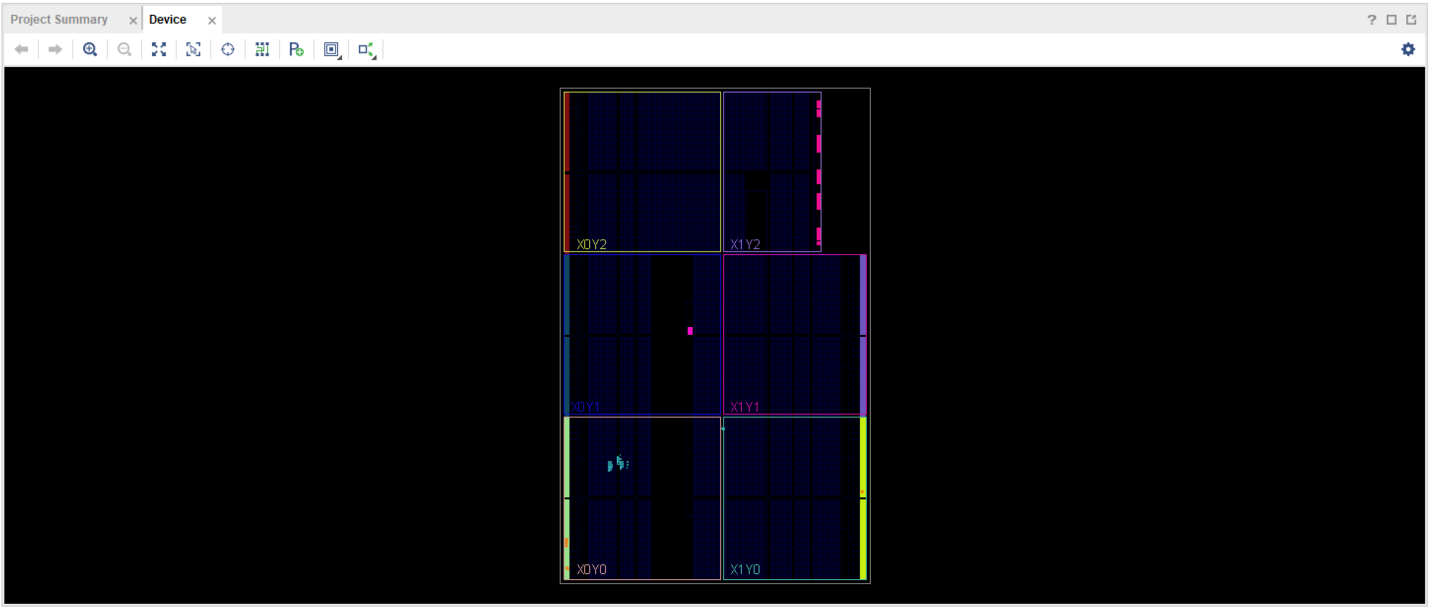
All user specified timing constraints are met.

Critical Path:



Name	Slack	Levels	Routes	High Fanout	From	To	Total Delay	Logic Delay	Net Delay	Requirement	Source Clock	Destination Clock	Exception
Path 1	6.261	2	3	1	RAM_inst/mem_reg/CLKBWRCLK	SPI_inst/MISO_reg/D	3.623	2.823	0.800	10.0	sys_clk_pin	sys_clk_pin	
Path 2	6.945	2	3	13	SPI_inst/counter_reg[3]/C	SPI_inst/counter_reg[0]/CE	2.673	0.901	1.772	10.0	sys_clk_pin	sys_clk_pin	
Path 3	6.945	2	3	13	SPI_inst/counter_reg[3]/C	SPI_inst/counter_reg[1]/CE	2.673	0.901	1.772	10.0	sys_clk_pin	sys_clk_pin	
Path 4	6.945	2	3	13	SPI_inst/counter_reg[3]/C	SPI_inst/counter_reg[2]/CE	2.673	0.901	1.772	10.0	sys_clk_pin	sys_clk_pin	
Path 5	6.945	2	3	13	SPI_inst/counter_reg[3]/C	SPI_inst/counter_reg[3]/CE	2.673	0.901	1.772	10.0	sys_clk_pin	sys_clk_pin	
Path 6	6.964	1	2	6	SPI_inst/rx_valid_reg/C	RAM_inst/mem_reg/WEA[0]	2.324	0.751	1.573	10.0	sys_clk_pin	sys_clk_pin	
Path 7	6.964	1	2	6	SPI_inst/rx_valid_reg/C	RAM_inst/mem_reg/WEA[1]	2.324	0.751	1.573	10.0	sys_clk_pin	sys_clk_pin	
Path 8	6.987	2	3	13	SPI_inst/FSM_onehot_cs_reg[2]/C	SPI_inst/data_reg_reg[0]/CE	2.631	0.875	1.756	10.0	sys_clk_pin	sys_clk_pin	
Path 9	6.987	2	3	13	SPI_inst/FSM_onehot_cs_reg[2]/C	SPI_inst/data_reg_reg[1]/CE	2.631	0.875	1.756	10.0	sys_clk_pin	sys_clk_pin	
Path 10	6.987	2	3	13	SPI_inst/FSM_onehot_cs_reg[2]/C	SPI_inst/data_reg_reg[2]/CE	2.631	0.875	1.756	10.0	sys_clk_pin	sys_clk_pin	

Implementation:



Reports | Design Runs | Power | Methodology | Timing

Design Timing Summary

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	5.913 ns	Worst Hold Slack (WHS):	0.085 ns	Worst Pulse Width Slack (WPWS):	4.500 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	88	Total Number of Endpoints:	88	Total Number of Endpoints:	42

All user specified timing constraints are met.

Utilization Report:

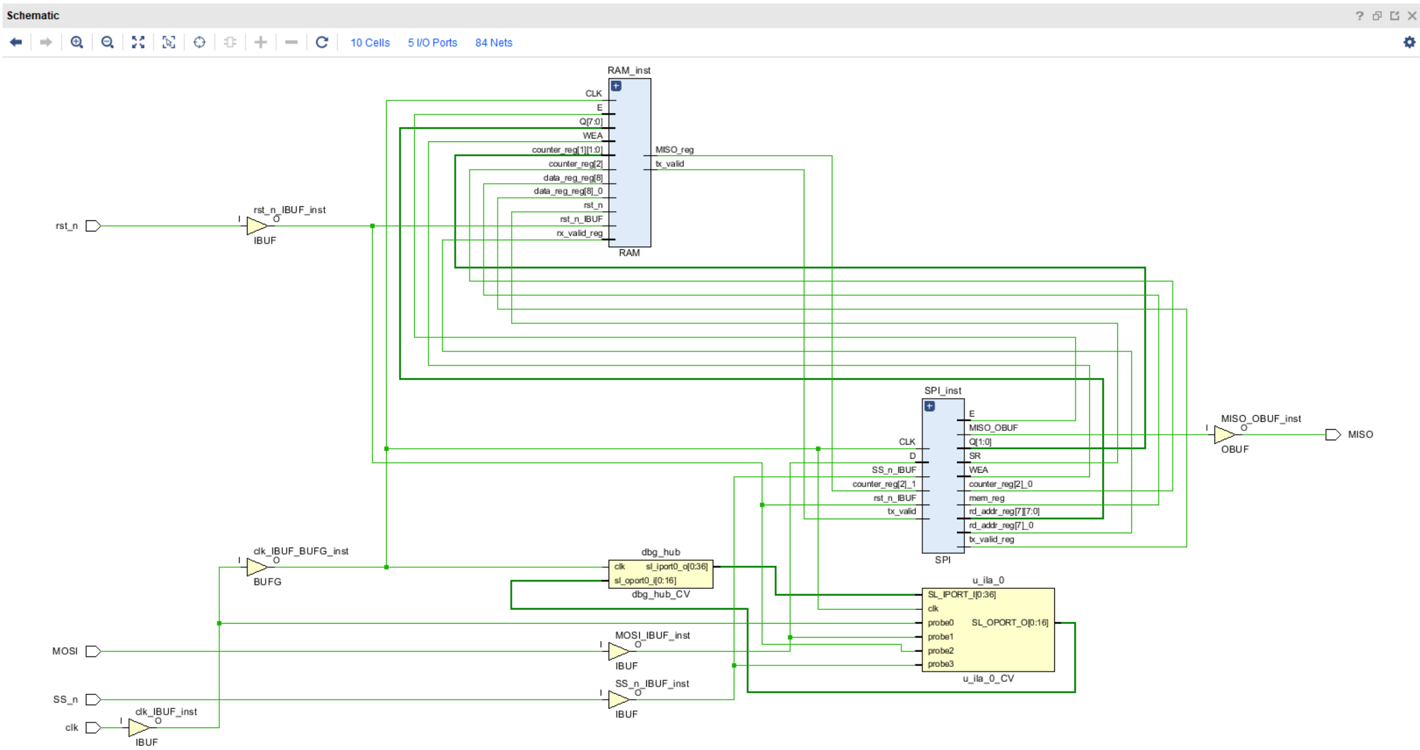
Design Runs | Power | Methodology | Timing | Utilization

Hierarchy

Name	Slice LUTs (20800)	Slice Registers (41600)	F7 Muxes (16300)	Slice (8150)	LUT as Logic (20800)	LUT Flip Flop Pairs (20800)	Block RAM Tile (50)	Bonded IOB (106)	BUFGCTRL (32)
SPI_RAM_TOP	38	39	1	21	38	12	0.5	5	1
RAM_inst (RAM)	3	17	1	6	3	0	0.5	0	0
SPI_inst (SPI)	35	22	0	20	35	11	0	0	0

gray:

Schematic:



Synthesis Report:

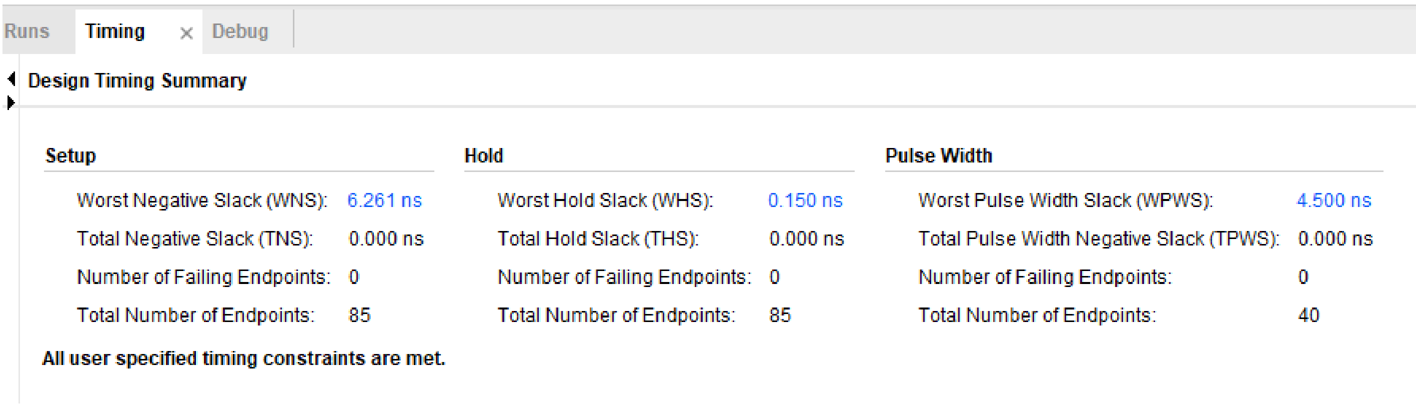

Schematic | synth_1_synth_synthesis_report_0 - synth_1

C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/project_1/project_1.runs/synth_1/SPI_RAM_TOP.vds

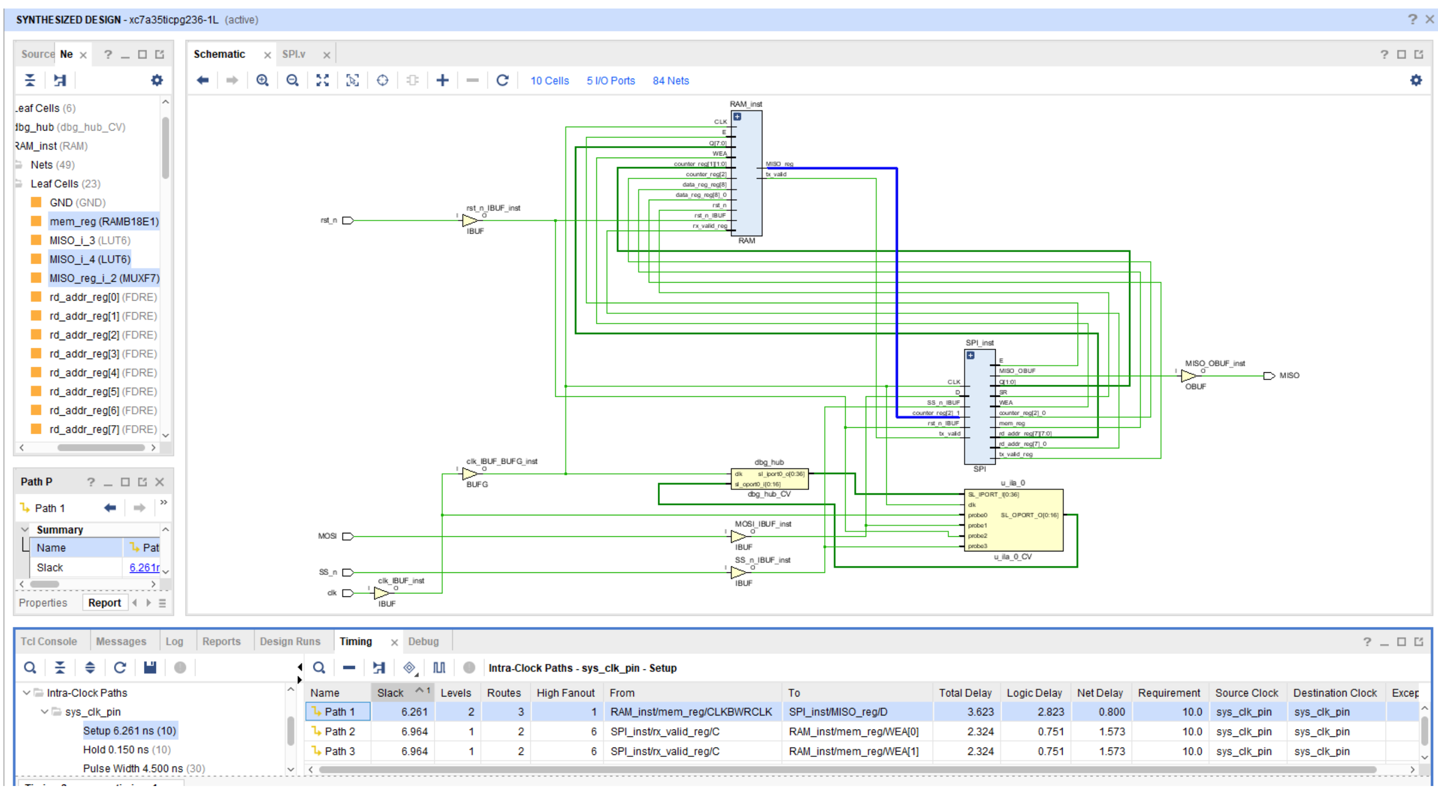
Read-only

```
    Parameter READ_ADD bound to: 3'b011
    Parameter READ_DATA bound to: 3'b100
INFO: [Synth 8-5534] Detected attribute (* fsm_encoding = "gray" *) [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/SPI.v:21]
INFO: [Synth 8-6155] done synthesizing module 'SPI' (1#1) [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/SPI.v:1]
INFO: [Synth 8-6157] synthesizing module 'RAM' [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/RAM.v:1]
    Parameter MEM_DEPTH bound to: 256 - type: integer
    Parameter ADDR_SIZE bound to: 8 - type: integer
INFO: [Synth 8-6155] done synthesizing module 'RAM' (2#1) [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/RAM.v:1]
INFO: [Synth 8-6155] done synthesizing module 'SPI_RAM_TOP' (3#1) [C:/Users/mahmo/OneDrive/Desktop/SPI_project/VivadoTest/SPI_RAM_TOP.v:1]
---------------------------------------------------------------------------------
Finished RTL Elaboration : Time (s): cpu = 00:00:02 ; elapsed = 00:00:02 . Memory (MB): peak = 410.172 ; gain = 153.047
---------------------------------------------------------------------------------

Report Check Netlist:
```

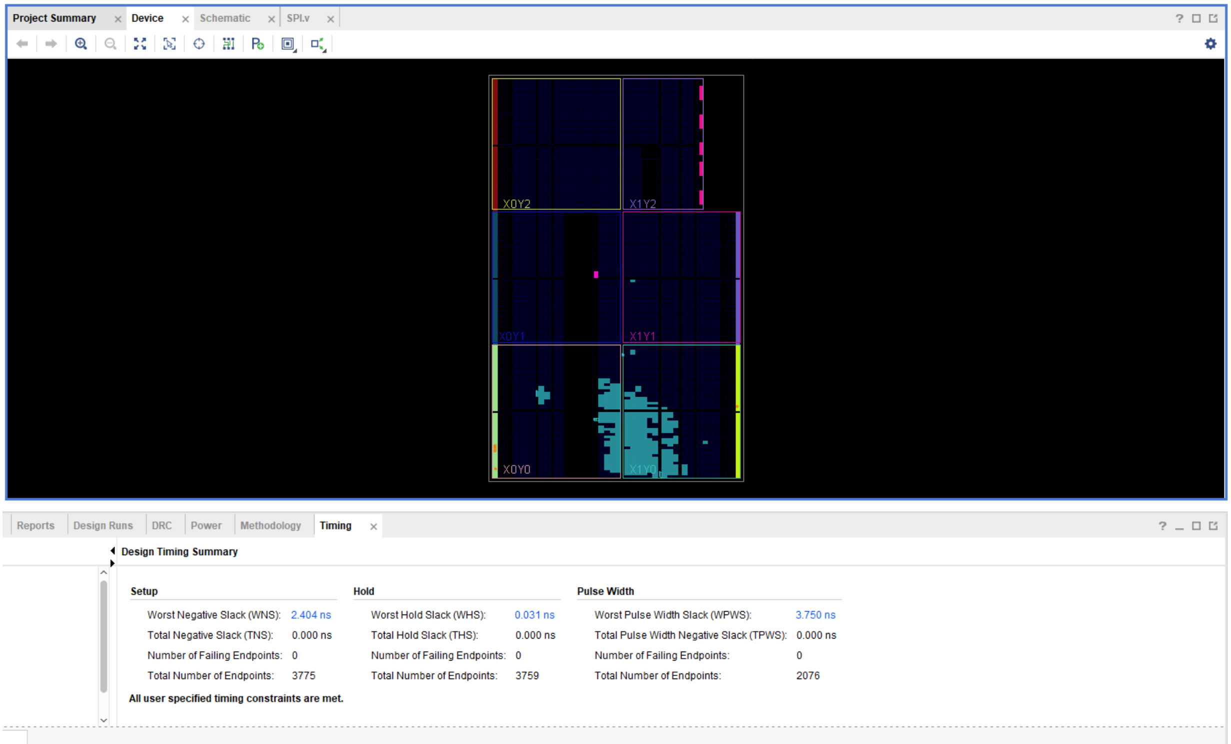
Timing Report:



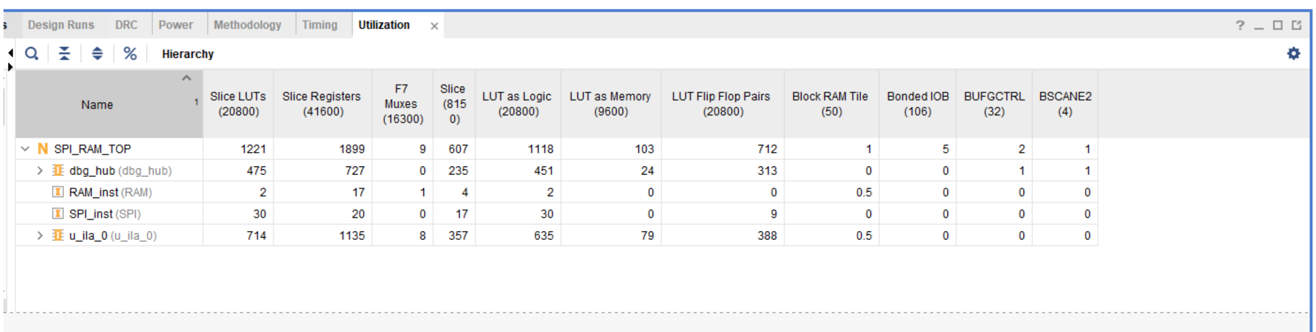
Critical Path:



Implementation:



Utilization Report:



Name	Slice LUTs (20800)	Slice Registers (41600)	F7 Muxes (16300)	Slice (815 0)	LUT as Logic (20800)	LUT as Memory (9600)	LUT Flip Flop Pairs (20800)	Block RAM Tile (50)	Bonded IOB (106)	BUFGCTRL (32)	BSCANE2 (4)
SPI_RAM_TOP	1221	1899	9	607	1118	103	712	1	5	2	1
dbg_hub (dbg_hub)	475	727	0	235	451	24	313	0	0	1	1
RAM_inst (RAM)	2	17	1	4	2	0	0	0.5	0	0	0
SPI_inst (SPI)	30	20	0	17	30	0	9	0	0	0	0
u_ila_0 (u_ila_0)	714	1135	8	357	635	79	388	0.5	0	0	0

Messages tab with no critical warnings or errors: